[Total No. of Question -04] [Total No. of Pages - 02]

G H Raisoni International Skill Tech University, Pune

S.Y. B. Tech (TERM-I)

CAE III- WINTER 2025 (2024 Pattern)

DISCRETE MATHEMATICS AND GRAPH THEORY (25BTBSL301)

(Time: - 01 Hour) (Max. Marks: 20)

Instructions to the students:

i) All questions are compulsory. ii) neat diagrams must be drawn wherever necessary.
iii) Figures to the right indicate full marks. iv) Use of scientific calculator is allowed
v) Assume suitable data, if necessary.

Q.No	Question	Marks	CO	BL
Q1.(a)	Let $A = \{1,2,3,4,5\}$, $R = \{(1,1),(1,2),(2,1),(2,2),(3,3),(3,4),(4,3),(4,4),(5,5)\}$ Find Transitive Closure of R by using Warshall Algorithm	[5]	1	3
b)	Show that A={1,2,3,5,6,10,15,30} is a POSET with divisibility on A. i.e. a divides b.	[5]	1	3
	OR			
Q 2. (a)	If $A = \{1,2,3,5,7,11,13\}$. And $R = \{(x,y) \mid x \text{ divides } y\}$. Find Relation R. and also draw a Hasse diagram	[5]	1	3
(b)	Determine whether the relation R={(1,2),(1,3),(3,1),(1,1),(3,3),(3,2),(1,4),(4,2),(3,4)} is reflexive, symmetric , transitive?	[5]	1	3
Q3.(a)	Prove that (Q,+, *) is Ring.	[5]	3	3
b)	*Let* $A = \{1,2,3,4\}$ *Find* $P(A)$*. i. e. Find Collection of all subsets of set A.* And draw a Lattice of $(P(A), \subseteq)$.	[5]	3	3
	OR			
Q 4. (a)	i. Define Integral domain with example. ii. Define zero divisors with examples.	[5]	3	2
(b)	Draw a Lattice of $(D_6, \mid)$. Where D_6 is set of all divisors of 6 for divisibility relation. Find composition table of meet and join	[5]	3	